

Marquee — AI Design Canvas

A viewing-discovery agent for a streaming catalogue · Aima Alakhume · BrainStation AI Certification, Final Project (Option 3)

UNOFFICIAL STUDENT CONCEPT · NOT AFFILIATED WITH DISNEY

01

Agent persona & task scoping

Persona. A catalogue specialist, not a chatbot. It speaks plainly, it never bluffs, and it would rather say what it could not do than answer something adjacent.

In scope. Turning a plain-sentence mood, constraint or occasion into titles that exist in this catalogue, with a reason attached.

Out of scope. Anything that is not finding something to watch. A request with no viewing intent gets one sentence and a redirect, not a guess.

Goal. Cut the time between opening the app and pressing play — measured by decisions reached, not clicks produced.

02

The users

A profile is **data, not a prompt**: watch history, weighted affinities over genres, tone tags and brands, and a set of negative weights. Switching profile changes the arithmetic, never the code path.

Same title, three profiles	Just me	Movie night	Date night
Tokyo Revengers anime, serialised, intense	12.4	2.8	0.6
Call It Love K-drama, subtitled, slow-burn	6.9	−3.0	4.7
Thor: Ragnarok superhero, comedic	1.0	8.2	1.6
Elemental romance, tender, animated	2.9	1.3	5.9

Kids mode is deliberately not a profile. It is a toggle any household can switch on, and a request can switch on for itself ("kid-friendly detective shows"). Safety that lives inside one persona is safety you lose the moment someone picks another.

03

System contract

Input	One sentence · the active profile · optionally a refinement of the last answer
Output	Exactly one of: ranked picks with grounded reasons · one clarifying question · a refusal naming the rule that refused
Invariants	Every title returned exists in the catalogue by id · at most two clarifying turns · no constraint is ever loosened silently
Stop	Picks delivered · question asked · refusal issued · catalogue exhausted after the published relaxation order

05

Knowledge stack & tools

- Catalogue** — 99 titles curated from Disney+ US: runtime, certification, genre, brand.
- Derived tone layer** — `energy` and `warmth` (1–5) plus tone tags, written once by a model reading each synopsis and audited by hand. This is what makes "comforting" a query.
- Profiles** — history, affinities, negative signals, hard filters.
- Tools** — `retrieve()` structured filter · `rank()` weighted scoring · `validate()` catalogue membership.
- Model, two narrow jobs** — intent parsing and tone re-ranking of a shortlist. It never selects from the full catalogue and never has the last word.
- Three tiers** — live model → cached answer → deterministic core. The interface always states which answered.

04

Logic & flow

1 PARSE Sentence → structured request; confidence scored	2 CLARIFY One question, or proceed and say so. Cap: 2 turns	
3 RETRIEVE Deterministic filter; hard filters first	4 RANK Weighted score, then optional model re-rank on tone	5 VALIDATE Membership check; invented ids dropped
6 EXPLAIN Reason from metadata + this profile's history	7 REFINE Re-enters at step 3 with locked constraints	
EXIT · REFUSES A household rule removes the whole request. It names the rule and offers the nearest thing it is allowed to offer.		
EXIT · RELAXES Nothing matches. Constraints are surrendered in a published order — runtime, year, format, genre — and the viewer is told which.		
EXIT · CATCHES A title the model invented is dropped before display; the step re-runs on what survived.		

06

Guardrails & failure handling

- Safety is structural.** The audience filter lives inside retrieval. "Ignore the parental settings" is not actionable, because no unsuitable title is ever a candidate.
- A rating is not an audience.** `kid_safe` is modelled separately from `certification`, because Abbott Elementary is TV-PG and is not a programme for a six-year-old.
- Grounding.** Ids are checked against the catalogue; anything invented is dropped, up to two retries.
- Honest degradation.** Every relaxation is named. A refusal says which rule refused.
- Locked refinements.** A constraint the viewer just set cannot be undone by the fallback path.
- Turn cap.** No clarification loop beyond two turns.
- Scope guard.** No viewing intent → one line, one redirect.
- No-model path.** Retrieval, ranking and every guardrail run locally, so the agent degrades to deterministic rather than to nothing.

07

Evaluation & optimisation

Twenty-five cases across happy path, ambiguity, empty results, child safety, prompt injection, out-of-scope, contradiction, refinement, negation, genre enforcement and vocabulary. Constraint satisfaction and grounding are checked mechanically on every case; taste fit is scored 1–5 by hand.

Run	Passing	What the run taught
v1	9 / 15	Two lexicon gaps; three design faults — a filter that only scored, a fallback that overrode an explicit instruction, a cache that outranked the user
v2	15 / 15	Category tags filter; refinements lock what they tighten; a stated mood halves the weight of standing taste
v7	25 / 25	Seven rounds. Negation contractions, a genre that moods could satisfy, a rating standing in for an audience, and a closed vocabulary that dropped words in silence

Full log, all seven rounds: `docs/iteration_log.md`.
Harness: `build/run_evals.mjs`.